Acute Recreational Noise Exposure Suppresses Behavioral but Not Metabolic Stress Responses in the Hairy Shore Crab (Hemigrapsus oregonensis)

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Abstract

Coastal marine environments face escalating noise pollution from shipping, construction, and recreational activity, yet the physiological consequences for near-shore invertebrates remain generally unstudied. We know that crustaceans can detect low-frequency vibrations through mechanoreceptors and may be sensitive to sounds produced by events such as shoreline festivals and floating concerts. Prior work demonstrated vibration-induced stress in *Carcinus maenas* and limited tolerance to shipping noise in *Hemigrapsus oregonensis*, but the effects of recreational EDM-style noise on Pacific Northwest intertidal crabs have not been investigated. Here we show that hairy shore crabs exposed to approximately 75 dB of low-frequency electronic dance music for two hours per week across three weeks showed longer righting times than control crabs, while respiration and hemolymph glucose showed no statistically significant treatment effect. Noise-exposed crabs had a median righting time of 0.94 s compared to 0.53 s in controls; two extreme outliers of 46.81 s and 56.24 s were additionally observed in the treatment group, suggesting severe disorientation in a subset of individuals which could represent a significant subset of individuals if we had more crabs. Visual trends in the resazurin fluorescence data consistently showed lower oxygen consumption in noise-exposed crabs across all three weeks despite non-significant p-values, and hemolymph glucose showed more variance in the treatment group. These findings suggest that behavioral impairment might precede or even occur independently of metabolic changes under acute recreational noise, and that nearshore invertebrate communities including Dungeness crab (*Metacarcinus magister*) may face underappreciated physiological risks from growing coastal recreational activity. Further investigation using larger sample sizes, longer exposure durations, and crustacean-specific assay protocols is warranted and the pilot study was a success.

Introduction

Washington State's Dungeness crab (*Metacarcinus magister*) fishery is in accelerating decline. South Puget Sound harvests peaked at 289,500 pounds in 2012 before collapsing to 9,500 pounds in 2017, and statewide catch in the 2024 to 2025 season declined 23% from the prior year and 44% from two seasons prior (Good, 2025; WDFW, 2026). The fishery faces compounding threats from the invasive European green crab (*Carcinus maenas*), ocean acidification, and habitat degradation (Jensen et al., 2002; Miller et al., 2016; Fisher et al., 2024). Reducing additional sources of preventable stress is essential for any recovery effort, and anthropogenic noise pollution, which is directly alterable through regulation, represents one such underexplored stressor.

Marine soundscapes have been altered by human activity, with underwater noise intensity said to be doubling each decade due to shipping, boating, and coastal development/construction (Jalkanen et al., 2022; Duarte et al., 2021). While impacts on marine mammals and fish are very well documented (especially the former), the physiological consequences for marine invertebrates are far less understood (Sole et al., 2023). Evidence is beginning to emerge Aimon et al. (2021) showed that *C. maenas* exposed to anthropogenic underwater vibrations showed elevated activity and antennae beat rates, showing that intertidal crabs can be noise sensitive. A meta-analysis by Davies et al. (2024) confirmed significant negative effects of sound on invertebrate behavior and physiology broadly, and Pani et al. (2026) documented behavioral and molecular stress responses in juvenile crustaceans under artificial noise.

The hairy shore crab (*Hemigrapsus oregonensis*) is a Pacific Northwest intertidal species with strong morphological and ecological similarities to Dungeness crab, including shared body plan, 10 jointed limbs, and an abdomen flap (WDFW, 2023). It is also a primary prey item for European green crabs, so its population health is directly tied to Dungeness crab ecosystem dynamics (Fisher et al., 2024). Birch et al. (2025) found that *H. oregonensis* showed limited physiological response to shipping noise, but the effects of lower-frequency, high-amplitude recreational noise remain untested. The objective of this study was to determine whether acute, repeated exposure to recreational EDM-style noise induces measurable physiological stress responses in *H. oregonensis*, measured via resazurin-based respiration, terrestrial righting time, and hemolymph glucose. We predicted that noise exposure would elevate all three indicators relative to controls.

Methods

Study Organism and Experimental Design

Twelve *H. oregonensis* collected from the Puget Sound were divided into six noise-exposed (treatment) and six control individuals. Selection was non-random to match body masses between groups plus gravid females were excluded to avoid reproductive metabolic variation. Each crab was tagged with a color-coded sticker fixed to the carapace with cyanoacrylate adhesive (yellow for control, blue for treatment) for individual tracking. Crabs were housed in separate aerated tanks at 35 ppt salinity and 10 degrees Celsius throughout the three-week experiment. Two control crabs died following molting events between weeks 1 and 2 and were excluded from all subsequent analyses, reducing the final sample size to 10.

Noise Exposure

Treatment crabs were exposed to approximately 75 dB of electronic dance music and techno (Techno Rave playlist, Magic Records via Apple Music) played from a JBL Go 4 Bluetooth speaker placed flush against the treatment tank for two hours once per week. Control crabs were simultaneously relocated to a separate acoustically isolated room. During the first hour crabs remained in home tanks; during the second hour crabs were transferred to resazurin well plates for respiration measurement.

Figure 1. Experimental setup during the first hour of noise exposure. The JBL Go 4 speaker was positioned adjacent to the treatment tank. Control crabs were housed in a separate room during this period. Shore crabs (Hemigrapsus oregonensis) are visible at the base of the tank.

Respiration, Righting Time, and Glucose Assays

Metabolic rate was measured with a resazurin fluorescence assay following Huffmyer et al. (2025). After one hour of noise exposure, all crabs were placed individually into wells of a 6-well plate containing 8 mL of resazurin working solution. Aliquots of 200 microliters were collected from each well and a paired blank at T0, T1 (30 min), and T2 (60 min), then transferred to a 96-well plate. Fluorescence was measured with a spectrophotometer and values were normalized by individual crab mass and blank-corrected. Following the resazurin assay and a two-stage saltwater rinse, righting time was recorded for each crab by timing the return from an inverted position. Outliers were identified using the Tukey method. In week 3 only, hemolymph glucose was measured from approximately 15 microliters of hemolymph extracted per crab, using the Cayman Chemical Glucose Colorimetric Assay Kit. All statistical analyses were performed in R version 4.5.3 and Microsoft Excel.

Results

Respiration

Resazurin fluorescence increased in both groups across all three weeks, confirming active respiration throughout the assay (Figs. 2 and 3). No significant difference in fluorescence was detected at any timepoint (T0: $p = 0.91$, T1: $p = 0.57$, T2: $p = 0.54$). Despite non-significance, control crabs consistently showed higher mean fluorescence than treatment crabs at both T1 and T2 in all three weeks, indicating greater oxygen consumption in the absence of noise. A paired t-test across all timepoints and weeks approached but did not reach significance ($t = -2.20$, $df = 8$, $p = 0.059$). Respiration increased nonlinearly in both groups, with greater acceleration from T1 to T2 than from T0 to T1.

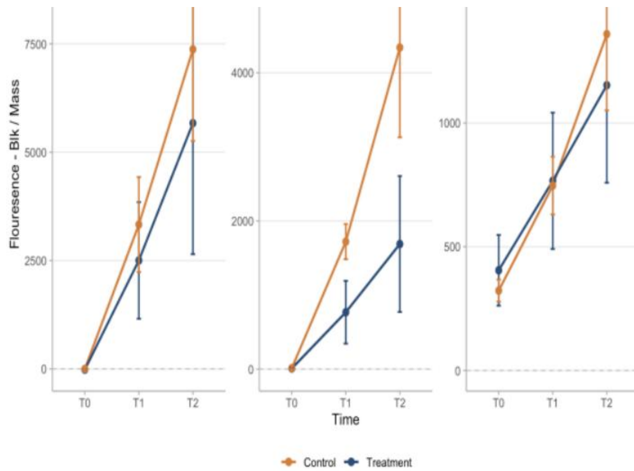

Figure 2. Resazurin fluorescence zeroed by blank and normalized by crab mass, shown separately for each of the three experiment weeks. Orange = control, blue = noise-exposed. Error bars are plus or minus one standard deviation. Y-axis scales differ between weeks.

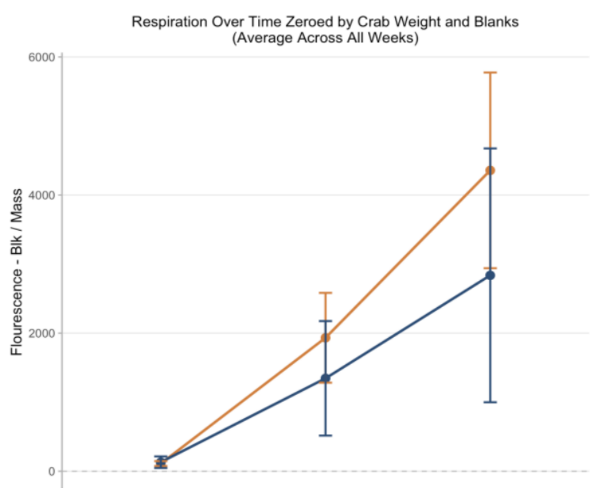

Figure 3. Mean resazurin fluorescence averaged across all three weeks at 30-minute intervals. Fluorescence zeroed by blank and normalized by mass. Orange = control, blue = noise-exposed. Error bars are the average standard deviation across all weeks.

Righting Time

Noise-exposed crabs had a higher median righting time (0.94 s) than controls (0.53 s) across all three weeks (Fig. 4). Two extreme values from the treatment group, 46.81 s and 56.24 s identified as outliers were also excluded from visualization. Including these outliers raised the mean treatment righting time to 6.79 s; excluding them, means were 1.23 s (SD = 0.92) for treatment and 1.06 s (SD = 1.29) for control. A t-test was non-significant (t = -0.40, df = 19.2, p = 0.69), though the consistent direction of the trend and the magnitude of the extreme outliers are biologically noteworthy.

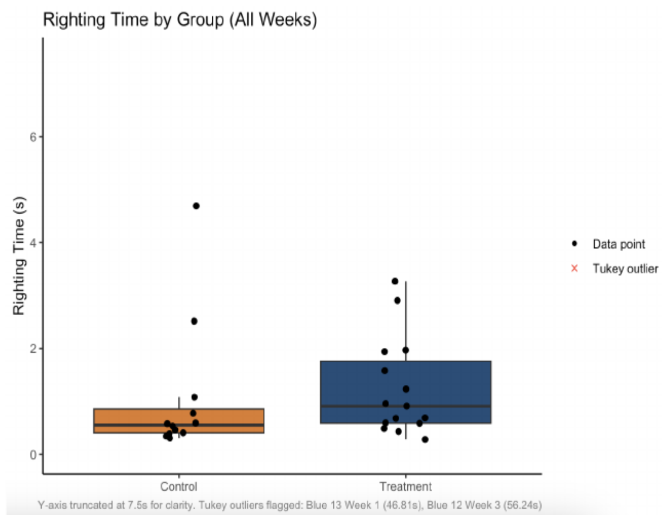

Figure 4. Terrestrial righting time in seconds pooled across all three weeks. Boxes show interquartile range, horizontal lines show median, whiskers extend to 1.5 times the IQR. Individual data points shown as filled circles. Two Tukey outliers from the treatment group (46.81 s and 56.24 s) were excluded from this visualization.

Hemolymph Glucose

Hemolymph glucose absorbance did not differ significantly between groups in week 3 (Fig. 5). Control crabs had a tight range of absorbance values centered near 0.10 nm. Noise-exposed crabs showed a substantially wider range, spanning from a minimum of 0.053 nm to a maximum of 0.227 nm, compared to a much narrower spread in controls.

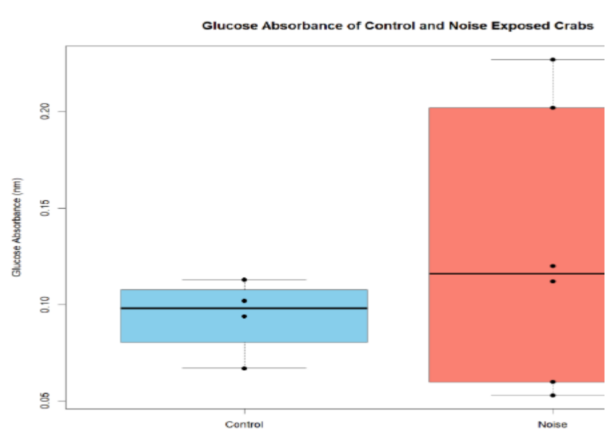

Figure 5. Hemolymph glucose absorbance from week 3 measured with the Cayman Chemical Glucose Colorimetric Assay. Blue = control, orange-red = noise-exposed. Boxes show interquartile range with median indicated.

Discussion

The most consistent finding was that noise-exposed crabs showed longer righting times than controls, with two extreme values suggesting severe acute disorientation in a subset of treatment individuals. Aimon et al. (2021) reported comparable locomotor impairment in *C. maenas* under anthropogenic vibrations, and the torpor-like behavioral suppression we observed aligns with crustacean stress response patterns described by Hapsari et al. (2025). Davies et al. (2024) similarly found that behavioral endpoints are more detectable than metabolic ones under acute noise exposure in invertebrates, consistent with our results.

The lack of significant metabolic responses may reflect partial tolerance to short-term noise in *H. oregonensis*, consistent with Birch et al. (2025), or may indicate that two hours of weekly exposure was insufficient to produce cumulative metabolic stress detectable at our sample size. Three molting events in the control group, two of which were fatal, likely elevated control respiration independently of noise treatment and reduced statistical power for detecting treatment effects. Cannibalism of dead molted crabs and associated water quality changes in the control tank during week 2 represent an important confounding factor. The glucose assay was additionally complicated by the use of a human blood kit on crustacean hemolymph, and the wider variance observed in the treatment group may partially reflect individual differences in molt stage, sex, and baseline stress tolerance rather than a uniform treatment response. Future studies should use crustacean-validated glucose assays and increase sample sizes substantially.

If the behavioral suppression observed here extends to wild populations, the implications for Pacific Northwest crab communities are significant. Noise-induced reduction in escape performance could amplify predation by *C. maenas* on *H. oregonensis* (Fisher et al., 2024), and the shared physiology between hairy shore crabs and Dungeness crabs means similar effects may occur in *M. magister*. Noise-induced suppression during spawning or larval settlement could compound the already severe population pressures on the Dungeness fishery (Good, 2025). We conclude that acute recreational noise at 75 dB disrupts behavioral performance in *H. oregonensis* more readily than metabolic function under the conditions tested here. Behavioral responses appear to manifest before measurable physiological changes, suggesting that righting time and similar behavioral assays may be the most sensitive indicators for future noise pollution research. Expanded studies with larger sample sizes, longer or continuous exposures, and direct testing on *Metacarcinus magister* will be necessary to evaluate whether recreational noise warrants inclusion in coastal noise ordinances and conservation planning for Puget Sound.

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